Loan Default Prediction

Exploratory Data Analysis & Predictive Modeling Report

1. Overview

This report presents a comprehensive analysis of a loan application dataset to predict loan default risk. The analysis consists of two major phases: exploratory data analysis (EDA) to understand structure, quality, and patterns in the data, followed by predictive modeling comparing seven machine learning models of increasing complexity. The goal is to identify the most effective model for classifying loan applicants as likely to default (class 1) or repay (class 0), given data about an individual's personal finances and credit history.

2. Dataset

The raw dataset contains 58,645 loan application records across 12 features, including the target variable (loan_status) and 11 variables relating to borrower demographics, personal finances/credit, and loan characteristics. There are no missing values or duplicate entries in the data. Features include:

- Borrower demographics: person_age, person_income, person_emp_length, person_home_ownership
- Loan characteristics: loan_amnt, loan_intent, loan_int_rate, loan_percent_income
- Credit history: cb_person_default_on_file, cb_person_cred_hist_length
- Target variable: loan_status

3. Exploratory Data Analysis (EDA)

3.1. Target Variable Class Imbalance

The target variable, loan_status, is a binary variable (0 = repaid, 1 = default) with a highly imbalanced distribution. Class 0 makes up 85.8% of the data, while class 1 makes up the remaining 14.2%, resulting in a repaid-to-default ratio of 6:1. This imbalance has significant implications for modeling: the simplest majority-class baseline, which predicts the majority class for every observation, would achieve over 85% accuracy while completely failing to identify any defaults. Because of this, accuracy is an insufficient metric to evaluate model performance on, and other metrics such as precision, recall, F1, and AUC were used throughout the analysis. Additionally, stratified sampling was used to preserve the class ratio across all data subsets.

3.2. Data Cleaning

Several features required cleaning to remove implausible values before the analysis and modeling stages. In total, 84 observations (0.14% of the original dataset) were removed, leaving the final cleaned dataset with 58,561 records.

person_age

The age variable ranged from 20 to 123. The maximum value (123) was isolated from the rest of the data by a 39-year gap from the second-highest value (84). This was a single observation, suggesting a data entry error or encoding rather than a true measurement. For this reason, the observation was dropped from the dataset.

person_emp_length

The employment length variable was interpreted as years of employment at a current job rather than over cumulative lifetime employment, based on the scatter of low employment length values across all age groups, including older applicants. A logical constraint was applied: employment start age (person_age - person_emp_length) must be at least 15 years old, consistent with US labor laws. 50 observations (0.09%) that violated this constraint were removed.

loan_percent_income

The loan percent income variable was cross-validated against the ratio computed directly (loan_amnt/person_income). Six rows showed discrepancies larger than 1.0, with the computed ratio in some cases exceeding 3x the annual income, which is implausible. These six observations were dropped.

cb_person_cred_hist_length

The credit history length variable suffered from a similar issue to the employment length variable: several observations show credit history beginning at extremely early ages for some individuals. A similar logical constraint was applied: the implied credit start age (person_age - cb_person_cred_hist_length) must be at least 17. A subset of 27 rows (0.045%) that violated this constraint was dropped from the dataset.

3.3. Univariate Analysis

Key distributional findings across individual features:

- Person_income and loan_amount both exhibited strong right-skewedness, with values ranging from the low thousands to 1.9M, with the 99th percentile at $175,000. Log transformations were applied for logistic regression; raw values were used for tree-based models.
- RENT (52%) and MORTGAGE (42%) dominate the person_home_ownership variable, while OWN (5%) and OTHER (0.15%) make up a minuscule proportion of the data.
- loan_intent is distributed relatively evenly across the six categories: EDUCATION (21%), MEDICAL (18.6%), PERSONAL (17%), VENTURE (17%), DEBTCONSOLIDATION (15.6%), and HOMEIMPROVEMENT (10.7%)
- loan_grade is heavily concentrated in grades A and B, with counts declining as grades worsen
- loan_amount exhibits a right-skewed distribution, with local spikes in loan amounts every $5,000. This is likely due to round-number clustering
- loan_int_rate follows a multimodal distribution, rather than a single, smoothed peak
- cb_person_default_on_file (prior default on record) is highly imbalanced, with 85% of the data falling into “N” and 14.8% in “Y”, exhibiting a similar imbalance to the target variable

3.4. Multivariate Analysis

Correlation analysis between numerical features and loan_status revealed that loan_percent_income and loan_amount are the variables most positively correlated with the target variable, though the magnitude of the correlations is fairly moderate (~0.3). Other numerical features show near-zero linear correlation with loan_status.

Categorical features showed stronger and more interpretable relationships with loan_status:

- loan_grade: the strongest predictor, with default rates climbing from 4.9% at grade A to 81.2% at grade G. Interest rates follow an identical gradient, ranging from 7% to 20% across all grades
- cb_person_default_on_file: applicants with a prior default have a nearly 3x higher default rate than those without (30% vs 11.4%)
- person_home_ownership: renters have the highest average default rate (22.3%), while homeowners have the lowest (1.4%)
- person_income: lower-income borrowers default more frequently, with the lowest income bin having an average default rate of 33.8% and the highest bin having a default rate of 3.8%
- person_age: the default rate remains stable across all age bins (11-16%), suggesting age may not have a meaningful relationship with loan_status

These findings point to loan_grade, loan_percent_income, loan_int_rate, and cb_person_default_on_file as the strongest predictors of defaulting.

4. Modeling Methodology

4.1. Data Partitioning

The cleaned dataset was split into three subsets using stratified sampling to preserve the class distribution across all partitions:

- Training set: 70% of the data, used to fit all models
- Validation set: 15% of the data, reserved for neural network hyperparameter tuning
- Test set: 15% of the data, used for final evaluation of all models

For tree-based models, 5-fold cross-validation within RandomizedSearchCV served as the validation strategy for hyperparameter tuning, making the validation set redundant for those models.

4.2. Feature Preprocessing

Two separate feature preprocessing pipelines were used, depending on the type of model:

Logistic Regression & Neural Network Pipeline

- Log transformation applied to person_income and loan_amnt to address right-skewed distributions
- Numerical variables were standardized using StandardScalar

- Ordinal encoding was applied to loan_grade (A=6 through G=0) to preserve the hierarchy of values
- One-hot encoding was applied to categorical variables, with one column dropped to serve as the baseline and prevent multicollinearity
- VIF analysis performed on numerical features; most features showed VIF < 5, with two features above 6.5, which were retained for baseline modeling

Tree-Based Pipeline

- Numerical features passed through without scaling or log transformation, as tree splits are not affected by feature scales
- Ordinal encoding was applied to loan_grade (A=6 through G=0) to preserve the hierarchy of values
- One-hot encoding was applied to categorical variables, without dropping a reference column (trees don't require a baseline)

4.3. Evaluation Metrics

Due to the class imbalance in the target variable (class 0: 85.8%, class 1: 14.2%), accuracy was not used as the primary evaluation metric. The following metrics were tracked across all models:

- Precision: proportion of predicted defaults that are actual defaults
- Recall: proportion of actual defaults that the model identified correctly
- F1 Score: harmonic mean of precision and recall; primary metric for model comparison
- AUC-ROC: model's ability to distinguish between classes across all thresholds
- Accuracy: reported for completeness, but treated as a secondary metric

In a lending context, recall is the most critical evaluation metric. A missed default (false negative) results in a loan being approved for a borrower who will ultimately default, resulting in financial losses. A false alarm (false positive) results in a loan being denied to a creditworthy borrower, a missed business opportunity, but no financial loss.

4.4. Class Imbalances

Several strategies were used across the various models to address the 6:1 class imbalance in the target variable. For logistic regression and decision trees, class_weight = 'balanced' was tested, which scales the loss function by the inverse class frequency. For XGBoost, scale_pos_weight was calculated as the ratio of negative to positive training samples (neg/pos), which directly instructs the model to upweight the minority class. For neural networks, the validation set was used to monitor per-epoch performance and identify the optimal stopping point.

5. Modeling Results

5.1. Model Comparison

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	90.5%	77.5%	47.0%	58.5%	0.903
Logistic Regression + Class Weights	82.3%	43.6%	83.6%	57.3%	0.903
Decision Tree	91.5%	69.9%	70.7%	70.3%	0.828
Decision Tree + Class Weights	91.9%	73.3%	68.0%	70.6%	0.819
Random Forest (Tuned)	94.9%	93.8%	69.3%	79.7%	0.941
XGBoost (Tuned)	93.5%	74.5%	82.6%	78.3%	0.959
Neural Network (Tuned)	94.6%	90.6%	69.0%	78.3%	0.928

5.2. Logistic Regression

The baseline logistic regression model achieved 90.5% accuracy and an AUC of 0.903, establishing a strong baseline. However, its recall of only 47% exposed a significant inability to detect actual loan defaults at the default classification threshold, correctly identifying only half of the actual defaults. Precision was 77.5%, yielding an F1 of 58.5%. Training and test metrics were closely aligned, indicating good generalization.

Introducing class_weight = ‘balanced’ significantly improved recall to 83.6%, but caused precision to fall dramatically to 43.6%, producing a high number of false positives and reducing accuracy to 82.3%. The AUC remained unchanged, and the F1 declined slightly due to the extreme precision-recall tradeoff.

5.3. Decision Tree

The simple decision tree improved F1 drastically, increasing to 70.3%, outperforming both logistic regression models on this metric. However, training metrics scored perfectly (1.0 across all metrics), indicating that the tree fully memorized the data it was trained on. Despite this, test performance was reasonable, suggesting the overfitting mainly reduced the model’s reliability rather than predictive validity on the test set. The ROC curve for the decision tree showed a sharp angular shape rather than the smooth curve produced by logistic regression. This is due to the nature of tree models, where they output hard probability estimates rather than probabilities across many thresholds. The AUC of 0.828 was noticeably lower than that of logistic regression (0.903).

Adding class_weight = ‘balanced’ to the decision tree yielded only marginal changes; precision increased slightly to 73.3%, recall decreased slightly to 68%, and F1 and AUC remained virtually unchanged.

5.4. Random Forest (Tuned)

The random forest model represented the largest single jump in performance in the analysis, with F1 improving significantly to 79.7%. Precision also increased dramatically to 93.8%, indicating that when the model predicted a loan to be default, it was correct more than 93% of the time, and produced only 57 false positives. This makes the random forest model the strongest for identifying defaults. However, the model’s low recall (69.3) resulted in 384 true defaults going undetected.

Hyperparameters were identified through RandomizedSearchCV, with 5-fold cross-validation scored on F1. The best parameters are as follows: n_estimators = 200, min_samples_split = 10, class_weight = None, and max_depth = None. The class_weight = None parameter suggests that the model performed better without explicitly penalizing the majority class, but rather relying on the averaging mechanism present in ensemble methods.

The AUC of 0.941 was an improvement over both the decision tree (0.828) and logistic regression (0.903), and the ROC curve was noticeably smoother than the decision tree’s angular shape. Training metrics scored marginally better than test metrics, which is to be expected with unseen data; it implies that the model generalizes well.

5.5. XGBoost (Tuned)

XGBoost emerged as the strongest performing model overall, achieving the highest overall recall (82.6%) and AUC (0.959) across all models. While its F1 score of 78.3% is marginally lower than the random forest (79.7%), the significant improvement in recall makes it the preferred model in a lending context where missed defaults lead to financial consequences.

The scale_pos_weight parameter was calculated as the ratio of negative to positive training samples, instructing XGBoost to upweight default cases during boosting, which simultaneously improved recall while maintaining a reasonable precision (74.5%).

The AUC of 0.959 is the highest of all the models, with a smooth ROC curve reflecting well-calibrated probability outputs across every threshold. Regularization parameters (min_child_weight, gamma) included in the grid search successfully prevented overfitting, and training and test metrics have little difference, indicating that the model generalizes well.

5.6. Neural Network (Tuned)

A comparison of nine neural network configurations was conducted, with varying hidden layer sizes ([none], [32], [64, 32], [128, 64]), activation functions (‘relu’, ‘tanh’), and optimizers (‘Adam’, ‘SGD’). Adam consistently outperformed SGD, and adding hidden layers steadily increased model performance, with the baseline no-hidden-layer configurations performing worst across all metrics.

The optimal configuration had the following parameters: hidden_layer_sizes = [128, 64], activation = ‘relu’, optimizer = ‘Adam’, and learning_rate = 0.01. This configuration was evaluated on the test set, achieving an accuracy of 94.6%, precision of 90.6%, recall of 69%, F1 of 77.4%, and AUC of 0.926.

This model’s precision score (90.6%) is the highest among all models evaluated, indicating that the neural network is rarely incorrect when flagging a default. However, with a recall of 69%, the model mirrors the random forest’s performance and misses 31% of actual defaulters. The training and test metrics have only small differences, indicating that the model generalizes well to unseen data.

6. Conclusion

This analysis evaluated seven machine learning models for loan default prediction, progressing from logistic regression through tree-based ensembles and neural networks.

Key findings:

- Class imbalance (85.8% repaid vs 14.2% default) required stratified sampling, alternative evaluation metrics (F1/AUC), and explicit imbalance handling across all models
- loan_grade is the strongest individual predictor of default risk, with default rates ranging from 4.9% (grade A) to 81% (grade G). loan_percent_income, loan_int_rate, and cb_person_default_on_file are also strong predictors
- Performance improved significantly as model complexity increased, with the baseline logistic regression achieving an F1 score of 58.5% to 79.7% with the random forest
- XGBoost is the recommended model for lending contexts where missed defaults lead to financial consequences, achieving the highest recall (82.6%) and AUC (0.958)